B V Raju Institute of Technology

4th Edition of R&D Showcase 2026

Department of CSE (Data Science)

A Trimodal Feature Fusion Framework for Explainable Brain Tumour Classification Using MRI Images

CSD-23-A-052

Communicated in Computer Methods and Programs in Biomedicine | SCI (Q1) | IF: 4.8

Abstract

- Developed an Explainable Hybrid CNN–Transformer Architecture for multi-class brain tumour classification from MRI scans, integrating local spatial feature extraction with global attention mechanisms.
- Designed to classify four categories: Glioma, Meningioma, Pituitary Tumour, and No Tumour using a multi-phase training strategy with Exponential Moving Average (EMA) for improved stability and generalisation.
- Incorporated Grad-CAM-based visual explainability and Structured Explanation Cards (SEC) to provide interpretable and clinically meaningful predictions.
- Achieved strong classification performance with high precision, recall, and F1-score, emphasising both diagnostic accuracy and model transparency for AI-assisted medical decision support.

Introduction

- Brain tumour diagnosis using MRI scans is critical for early treatment, but manual interpretation is time-intensive and subject to variability.
- Existing deep learning models achieve high accuracy but often lack interpretability, limiting clinical trust and adoption.
- CNNs effectively capture local spatial features, while Vision Transformers model global contextual relationships; however, individually, they have limitations.
- To address these challenges, this work proposes an Explainable Hybrid CNN–Transformer framework that combines performance, robustness, and interpretability for multi-class brain tumour classification.

Objectives

- Develop a hybrid CNN–Transformer architecture for accurate multi-class brain tumour classification from MRI scans.
- Integrate both local feature extraction and global attention mechanisms to enhance discriminative learning across tumour categories.
- Improve training stability and generalisation using a multi-phase training strategy with Exponential Moving Average (EMA).
- Incorporate model explainability techniques (Grad-CAM and Structured Explanation Cards) to provide clinically interpretable predictions.
- Achieve high precision, recall, and F1-score while maintaining robustness across all tumour classes.

Methods and Techniques

The overall methodology for multi-class brain tumour classification from MRI scans is summarised in the following algorithm.

Proposed Brain Tumour Classification Framework

Step 1: Dataset Collection
- Collected a labelled brain MRI dataset containing four categories: Glioma, Meningioma, Pituitary Tumour, No Tumour.
- Organised for supervised multi-class classification.

Step 2: Data Preprocessing
- Resize MRI images to a uniform input dimension.
- Normalise pixel intensity values.
- Apply data augmentation (rotation, flipping, scaling).
- Split the dataset into training, validation, and testing sets.

Step 3: Hybrid Model Development

Designed a hybrid deep learning model combining:
- ResNet50 for extracting local spatial features
- Swin Transformer for capturing global contextual information
- MobileViT for learning efficient hybrid representations

Step 4: Feature Fusion & Classification
- Extract feature embeddings from all three models.
- Perform feature-level fusion (concatenation).
- Pass fused features through fully connected layers.
- Apply Softmax activation to generate a final four-class prediction.

Step 5: Training Strategy
- Compute multi-class cross-entropy loss.
- Update model weights using backpropagation.
- Maintain an Exponential Moving Average (EMA) for stable learning.
- Train in multiple phases to improve convergence.

Step 6: Evaluation & Explainability
- Evaluate using accuracy, precision, recall, and F1-score.
- Generate a confusion matrix for class-wise analysis.
- Apply Grad-CAM to visualize tumour localisation.
- Present predictions using Structured Explanation Cards (SEC).

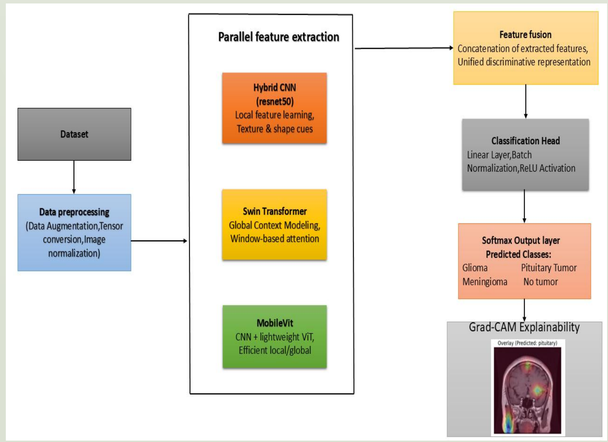

Figure 1: Architecture of the proposed methodology.

Results

1. Qualitative Results
- The proposed HybridTriModel achieved 99.54% accuracy under 10-Fold Cross-Validation with low standard deviation (±0.30), demonstrating stable and consistent performance.
- Across multiple evaluation strategies (70/30, 80/20, 5-Fold, 10-Fold), the model maintained high generalisation capability.

Table 1: Overall accuracy comparison of different models under various training and validation strategies

Model	70/30	80/20	5-Fold (± std)	10-Fold (± std)
CNN	94.26	92.95	93.83 ± 1.59	94.74 ± 0.52
CNN+Swin Hybrid model	98.81	98.36	98.99 ± 0.72	99.18 ± 0.59
Proposed Trimodal	98.77	99.36	99.10 ± 0.30	99.54 ± 0.30

Table 2: Class-wise precision, recall, and F1-score of the proposed trimodal model under different evaluation settings

	Proposed 70/30 Split			Proposed 80/20 Split		
Class	Precision	Recall	F1-score	Precision	Recall	F1-score
Glioma	99.59	98.97	99.28	100.00	99.07	99.53
Meningioma	98.75	96.15	97.44	99.39	99.09	99.24
No Tumour	97.56	99.83	98.68	99.50	99.75	99.63
Pituitary	99.43	99.81	99.62	99.15	100.00	99.58

	Proposed 5-Fold			Proposed 10-Fold		
Class	Precision	Recall	F1-score	Precision	Recall	F1-score
Glioma	99.69	100.00	99.85	100.00	100.00	100.00
Meningioma	99.39	98.48	98.93	99.94	99.21	99.57
No Tumour	99.50	100.00	99.75	99.75	99.60	99.57
Pituitary	99.43	99.43	99.43	99.10	99.94	99.57

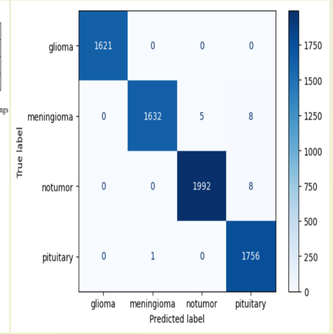

Figure 2: Results of the proposed model across different evaluation settings & confusion matrix

- The Confusion Matrix (CM) shows dominant diagonal values, confirming that most samples were correctly classified.
- Very minimal off-diagonal entries were observed, indicating low inter-class confusion and strong class separation.

2. Clinical Validation Results

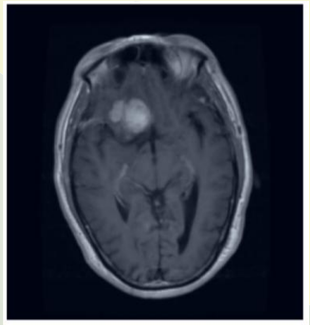

Figure 3: Input MRI Image

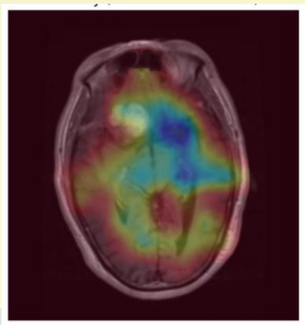

Figure 4: Resulted Output MRI Image

	SEC Section	Attribute	Value
0	Prediction Summary	Predicted Class	Meningioma
1	Prediction Summary	Predicted Index	1
2	Confidence Analysis	Prediction Confidence	0.999863
3	Confidence Analysis	Top-1 vs Top-2 Margin	0.999791
4	Attention Evidence (Grad-CAM)	Mean Attention Strength	0.099084
5	Attention Evidence (Grad-CAM)	Max Attention Strength	1.0
6	Attention Evidence (Grad-CAM)	Active Attention Area Ratio	0.059451
7	Decision Reliability	High Confidence Decision	True
8	Decision Reliability	Clear Class Separation	True

Figure 5: Structured Explanation Card (SEC)

- Grad-CAM heatmaps successfully localised tumour regions in MRI scans, focusing attention on abnormal tissue areas.
- The highlighted regions correspond to visually distinguishable lesion areas, supporting meaningful feature learning.
- Structured Explanation Cards (SEC) provide prediction summaries including predicted class, confidence score, and reliability indicators.
- The integration of visualisation and structured outputs enhances interpretability, supporting potential AI-assisted clinical decision support applications.

Table 3: Year-wise Comparison of Brain Tumour Classification Methods with Reported Performance Metrics

S.No	Authors	Year	Methodology	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
1	Seetha and Raja [2]	2018	CNN with small kernels	97.50	–	–	–
2	Sultan et al. [4]	2019	Deep CNN for multi-class classification	96.13	96.10	96.10	96.10
3	Jia and Chen [21]	2020	FAHS–SVM framework	98.51	–	–	–
4	Divya et al. [26]	2020	ResNet50 transfer learning	98.67	–	–	–
5	Saleh et al. [15]	2020	CNN-based transfer learning	98.75	–	–	–
6	Biswas and Islam [17]	2021	Hybrid CNN–SVM model	98.55	–	–	–
7	Majib et al. [31]	2021	VGG-SCNet stacked classifier	99.20	99.20	99.10	99.20
8	Nallamolu et al. [27]	2022	CNN-based multi-class classification	94.72	–	–	–
9	Vankdothu and Hameed [14]	2022	RCNN with GLCM features	95.17	–	–	–
10	Nayak et al. [16]	2022	Dense EfficientNet CNN	98.78	–	–	–
11	Sujatha and Rao [25]	2023	DenseNet201-based CNN	88.00	–	–	–
12	Hossain et al. [8]	2023	Vision Transformer ensemble	96.94	–	–	–
13	Ferdous et al. [23]	2023	LCDEiT vision transformer	98.11	–	–	–
14	De Benedictis et al. [3]	2024	TDA with Tucker decomposition	97.28	–	–	–
15	Guluwadi et al. [10]	2024	ResNet50 with Grad-CAM	98.52	98.50	98.50	98.50
16	Abdelhaliem et al. [19]	2025	Multimodal attention network	92.86	92.86	80.00	85.90
17	Dintakurthi et al. [18]	2025	CNN + Transformer hybrid	97.80	–	–	–
18	Hasan et al. [7]	2026	Attention-based explainable DL	97.00	97.00	97.00	97.00
19	**Proposed HybridTriModel**	**2026**	**ResNet50 + Swin Transformer + MobileViT**	**99.54**	**99.69**	**99.69**	**99.90**

Figure 6: Comparison of Brain Tumour Classification Methods with Reported Performance Metrics

Conclusions

This project proposed an explainable hybrid deep learning framework for multi-class brain tumour classification using MRI images by integrating ResNet50, Swin Transformer, and MobileViT through feature-level fusion. The model achieved 99.54% accuracy under 10-fold cross-validation with low variance, demonstrating stable and robust performance across evaluation strategies. High precision, recall, and F1-scores with minimal misclassification confirmed strong class separation. In addition to high predictive accuracy, Grad-CAM visualisations and Structured Explanation Cards enhanced transparency by highlighting tumour regions and providing confidence-based decision insights. Overall, the framework effectively combines performance and interpretability, showing strong potential for AI-assisted clinical decision support in medical imaging.

References

[1] D. M. Toufiq, A. M. Sagheer, H. Veisi, "A Review on Brain Tumour Classification in MRI Images," *Turkish Journal of Computer and Mathematics Education*, 2021.

[2] J. Seetha, S. S. Raja, "Brain Tumour Classification Using Convolutional Neural Networks," *Biomedical & Pharmacology Journal*, 2018.

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SDGs: 3, 9

SDG 3 – Good Health and Well-Being

This project supports early detection of brain tumours using AI-based MRI analysis. It helps improve diagnosis accuracy and supports better healthcare outcomes.

SDG 9 – Industry, Innovation, and Infrastructure

The project promotes innovation in medical technology by using advanced deep learning and explainable AI methods for healthcare applications.

Author Details

Name :	Roll No:
Kalali Sai Ram Goud	23211a6745
Lingampet Anjali	23211a6762
Gullakunta Prashanth	23211a6734

Guide Name:
Dr. Ramdas Kaplia

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